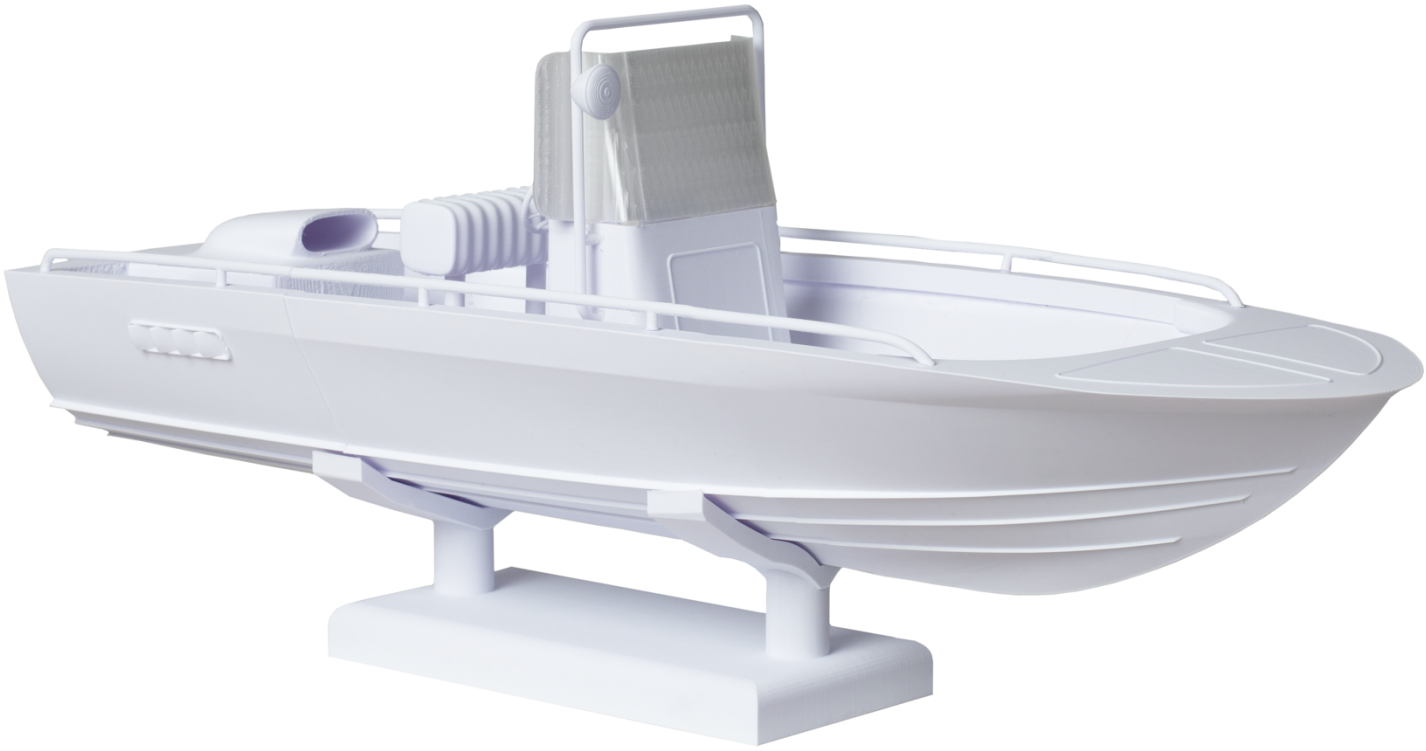
Deciphering and applying design cues-

The birth of the Mallard 24



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Introduction

A products design can be compared to an onion. It is composed of multiple layers of shapes, colors and proportions that combined create a united composition. However, in this composition of layers, a few fundamental design elements stand out and give the product its distinct appearance. These fundamental design elements can be referred to as the products design DNA or core design cues.

The Curtiss P-40 Warhawk, is an American airplane that was active during the second world war. With its sleek design, multiple exhaust pipes and over-dimensioned air intake, the P-40's design is recognizable and iconic.

The center console boat is to the contrary of the P-40, a recreational means of water transportation. With its practical features, open layout and reasonable price point, the center console boat has become a favorite boat among enthusiasts, fishermen and families.

With a starting point in published research and interviews, the aim of this thesis is to investigate three different design cue theories and thereafter test their relevance. The aim is to answer: Are design cue theories accurate and can they be applied to design a P-40 inspired center console boat.

Assumptions & restrictions

As the goal of this thesis is to investigate current design cue theories and test their applicability in practice, the P-40 was chosen as the doner product. The main reason for this is the P-40´s iconic and evident design cues. The distinct design cues make them graspable which facilitates in the deciphering and application process.

The choice of a center console boat as the subject product was based on two primary factors. The first factor is similarity between boats and airplanes. Both products are modes of transportation, both have engines and both have similar main fuselages. Although these similarities might not put the design theories to maximal stress, they facilitate the initial exploration of the theories. The second reason for using a boat as subject, is personal interest. The knowledge and interest in both airplanes and boats give motivation and context to conduct this research. Personal interest helps understand necessity and functionality, but also motivates to thoroughly analyze and implement explored theories.

In order to facilitate the understanding of elements discussed, a few terms can be explained. The center console boat has been given the name Mallard 24. The reason for this is that the mallard is a water bird that is nimble and reasonably fast. The number 24 stands for 24-feet or 7,2 meters which is the length of the boat. Other abbreviations used are P-40 for the airplane and CMF to describe color, material and finish of the product.

As for esthetic limitations, only one was taken into consideration. Although the P-40´s design is unique, its color scheme and painted symbols make it stand out from the crowd. In particular its painted shark mouth on the air intake makes it immediately identifiable. As these symbols are strongly related to the P-40 and can be considered a brand logo, they were chosen not to be taken into account. The reason for this restriction, is because the aim of this thesis is to explore design ques and not to transfer obvious attributes to achieve resemblance.



[1]

Finally, in order to put the Mallard into context, it was given a target audience. The Mallard was designed as a multipurpose boat for Swedish intercoastal use. The target audience is predominantly males age 30-60 who enjoy performance and classic design. They use the Mallard as a mean of transportation with their family on weekends and recreational activities such as waterskiing. The Mallard is designed to be driven but also as a status symbol. It is expected to be used as a classic sports car. That is to say; for pleasure rides on sunny days.

[1] Image nr 1. Killian, Curtis P-40M Warhawk.

Theory

In order to achieve a deeper understanding of design cue theory and its applicability, three different theories are explored. Firstly, visually driven design cue theory is explored which can be considered a traditional method for transferring design cues. Secondly, gender driven design cue theory is explored which can be considered as a radical design method. Thirdly, a holistic design theory is explored which can be considered a pragmatic approach to deciphering and applying design cues.

Visually driven design Cues

In the research article, "*It looks like a Toyota: Educational approaches to designing for visual brand recognition"*, the author Toni-Matti Karjalainen does an in-depth experiment to map what design cues are and how they can be transferred from one product to another. With the help of design students from Chalmers University, Karjalainen analyses design cues from several car brands and later applies them to household appliances such as watches, mp3 players and thermoses.[2]

In order to decipher design cues, Karjalainen categorizes design elements as explicit or implicit. With explicit design cues, Karjalainen means visual references that are recurring throughout the entire product range. This can be shapes, proportions or any design element that is reoccurring. For instance, Karjalainen sees that a V-shaped bonnet and distinctively carved backlights are explicit design cues that are consistent in Volvos cars.[3]

On the contrary, implicit design cues are combinations of references that are not evident at first glance. Instead, implicit design cues are value based, meaning that they evoke a sensation or association. In the case of Volvo, implicit design cues are related with Volvos core values being safety, dynamism and Scandinavian design heritage.[4] This can be illustrated with round

[2] Karjalainen, It looks like a Toyota: Educational approaches to designing for visual brand recognition, pp 1.
[3] Karjalainen, It looks like a Toyota: Educational approaches to designing for visual brand recognition, pp 3.
[4] Karjalainen, It looks like a Toyota: Educational approaches to designing for visual brand recognition, pp 4.

corners, sober colors and natural materials that tend to evoke the feeling of quality and safety. However, Karjalainen states that once these design cues are used repeatedly, they slowly move from implicit to explicit as they become visual cues that can be quickly recognized. With this said, there is not always a clear line between what is explicit and implicit. They tend to melt into each other depending on how obvious and reoccurring they are.[5]

One of Karjalainens key findings when transferring design ques from automotive brands to household appliances, was the difficulty of transferring explicit design cues. The reason for this is that these cues tend to be category-specific and context sensitive. With this Karjalainen means that transferring explicit design cues such as a car's grill, large windows or distinct shoulder-line to another product range can give a forced appearance as the design cue is not relevant for the household appliance.[6] Furthermore, Karjalainen describes that the test group was very efficient in transferring the value based implicit design cues from automotive brands to appliances, but not as effective when transferring explicit design cues.[7]

A final pointer that Karjalainen presents is the importance of balance when transferring design cues from one product to another. He states that transferring too many explicit and implicit design cues can result in a feeling of the product being overworked and unbalanced. He means that the key to a successful design transfer is to balance familiarity with novelty. [8] It is in this instance that the designers experience and understanding of composition is crucial.

Gender driven design Cues

In Karin Ehrnberger, Minna Räsänen and Sara Ilstedt research article: "Visualising Gender Norms in Design: Meet the Mega Hurricane Mixer and the Drill Dolphia", they use an unorthodox approach to deciphering product design cues. Instead of solely looking at the esthetics of a product and deciphering them by shapes and colors, they look at design through a gender perspective. They mean that a products design can be seen as a reproduction of

[5] Karjalainen, It looks like a Toyota: Educational approaches to designing for visual brand recognition, pp 10.
[6] Karjalainen, It looks like a Toyota: Educational approaches to designing for visual brand recognition, pp 10.
[7] Karjalainen, It looks like a Toyota: Educational approaches to designing for visual brand recognition, pp 10.
[8] Karjalainen, It looks like a Toyota: Educational approaches to designing for visual brand recognition, pp 3.

gender norms. Products thereby carry significant design cues that differ from each other depending on if the end consumer is male or female.[9]

According to Ehrnberger et al. products specifically designed for women tend to be round, clean and organically shaped. Furthermore, they tend to be brightly colored and often have floral related patterns. In contrast, products that are design for men tend to have complex and angular shapes. The color scheme is often darker and have a mechanical esthetic promoting excitement and performance.[10]

The combination of language and design indicators can clearly be seen in the design and packaging of irons. Although the technical specifications of both irons are similar, they convey completely different images. The iron design for males is dark, angular and highlighted with chrome and red indicators. While the iron designed for females is brightly colored, has rounded corners and a larger filler cap. Besides that, the male iron is labeled with words such as power, speed and boost, while the female iron is labeled with extra-long cord, soft glid and drop-stop system.[11]



[12]

Ehrnberger et al. refer the gender differentiation between the two irons as the principle of separation. This means that you have performance design cues on one end of the scale and helpfulness associated design cues on the other end. Performance design cues would be promoted towards males, while helpfulness would be promoted towards females. In this

[9] Ehrnberger et al. Gender Norms in Design: Meet the Mega Hurricane Mixer and the Drill Dolphia, pp 86.
[10] Ehrnberger et al. Gender Norms in Design: Meet the Mega Hurricane Mixer and the Drill Dolphia, pp 88.
[11] Ehrnberger et al. Gender Norms in Design: Meet the Mega Hurricane Mixer and the Drill Dolphia, pp 91.
[12] Image nr 2. Ehrnberger et al. Clothing iron.

sense, it is not only the actual form of the object that gives the product its design character, but instead it is whether the design elements are portrayed as performance- or helpfulness related.

One major conclusion reached by Ehrnberger et al. research, is the acknowledgement that design cues don't have to be categorized solely by traditional metrics like shapes and proportions.[13] Instead, Ehrnberger et al. suggest that when analyzing design cues with the intention of transferring them to another product, the question designers should be asking themselves is how performance or helpfulness associated is the donor's design? Once this question has been answered, a similar degree of e.g. performance indicating design cues can be applied on the subject product.[14] This means that no exact design element needs to be transferred from doner to subject, but rather the same level of performance vs helpfulness sensation in order to achieve resemblance between the products.

Ocke Mannerfelt – Holistic design approach

Ocke Mannerfelt is one of Sweden's most reputable boat designers. Still active at the age 85, Mannerfelt has designed over 200 boat models. In contrast to Ehrnberger et al. and Karjalainen, Mannerfelt has a more holistic approach when extracting the essence from a product and transferring it to a boat design. Mannerfelt starts with three basic principles: firstly, he looks at the esthetic composition of the donor product. Secondly, he has a tight dialog with customers to evaluate his thoughts and thirdly, he always has production costs in mind.[15] The question he always returns to is: will his design be able to be manufactured at a reasonable cost? Mannerfelt uses these guidelines when designing, but also when deciding which design cues to incorporate in the new design. [16]

In addition to his cost-related mindset when designing boats, Mannerfelt states that the ability to design an attractive boat is something that is learned through practice and experience. He argues that the esthetic eye is a trait an individual is born with and is trained with repetition.

[13] Ehrnberger et al. Gender Norms in Design: Meet the Mega Hurricane Mixer and the Drill Dolphia, pp 96.
[14] Ehrnberger et al. Gender Norms in Design: Meet the Mega Hurricane Mixer and the Drill Dolphia, pp 96.
[15] Mannerfelt, Design cue interview, 15/02/ 2025.
[16] Mannerfelt, Design cue interview, 15/02/ 2025.

With this said, Mannerfelt does not have a set of rules he uses when deciphering a design, he just gets to work and tries to identify traits that he thinks are valuable to transfer.[17]

Mannerfelts advises one final method to use when deciphering and applying design ques. He suggests to find a maximum of three lines that are significant for the donor products design and use those as a starting point. He means that when transferring design cues, the designer should simplify as much as possible. The goal is to simplify the donors design to a degree that it is recognizable with as few lines as possible. Once these lines are evident, it is easier to add new design elements but still achieve recognition to the donor product. He accentuates that length, width and height tend to be key lines to have in mind when deciphering a product's design.[18]

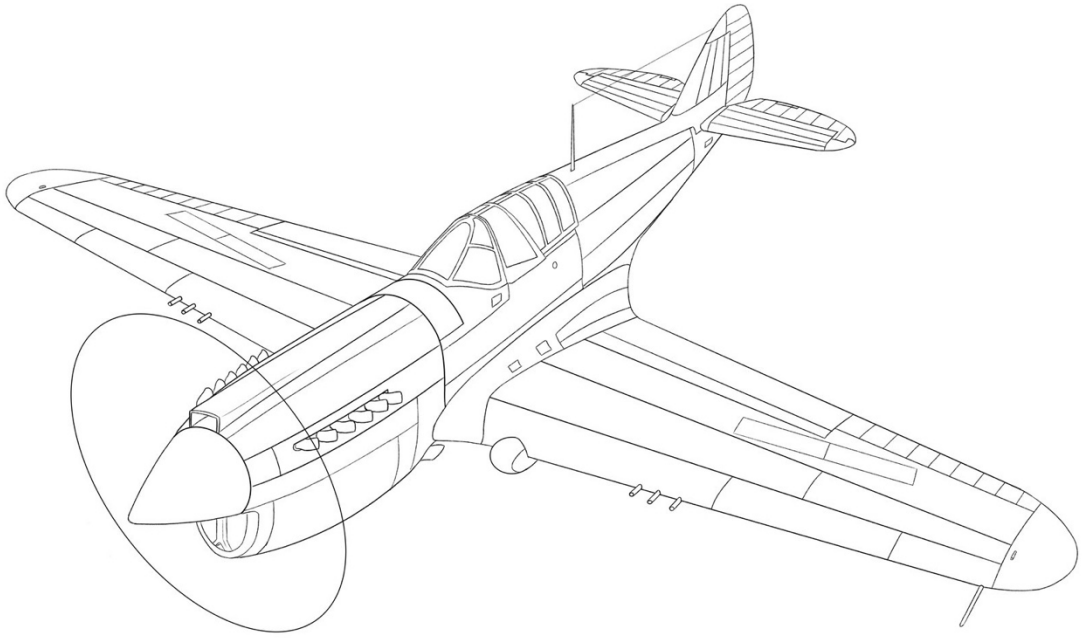
[17] Mannerfelt, Design cue interview, 15/02/ 2025.
[18] Mannerfelt, Design cue interview, 15/02/ 2025.

Analysis

Analysis from a visual perspective

When analyzing the P-40 Warhawk with Karjalainens method of visual recognition, both explicit and implicit design cue theory are applied. Interesting to note is how explicit and implicit design cues are interconnected. Just as Karjalainen suggested, the combination of explicit design cues are what create emotions that are experienced as implicit design cues.

Design elements of the P-40 identified as explicit include; enginge/fusalage ratio of 1:3, hard leading edges and round trailing edges, areo dynamic form, gemometric layout and broad shapes that taper towards the end . However, the most prominent explicit design cues acknowleged are the large air intake, multiple exahust pipes, and springy convex lines throughout the plane.



[19]

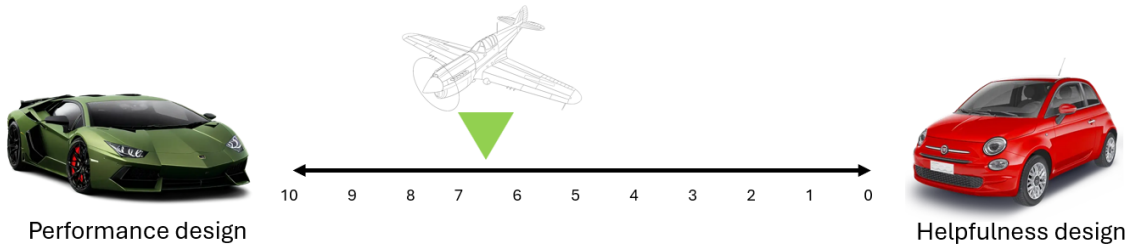
These explicit design cues create an emotinal sensation that is transferred to the P-40's implicit caracteristics. The P-40 is experianced as a fast, sturdy and relaiable object. It can be

[19] Image nr 3. Murphy, P-40 three quarter view.

compered to a flying comet that powers itself through the air with a tapering tail behind. The smooth surfaces, compact composition and long convex lines further more acentuate the feeling of the P-40 being fast and manoeuvrable. The multiple stumpy exahust pipes and the large air intake give the P-40 a feeling of raw preformance that promotes the feeling of speed and brute power. Finally, the equal length of wingspan vs fusalage gives a compact feeling wihich evokes a sensation of being manoeuvrable and nimble. To summarize, the P-40´s most prominent implicit design cues is a feeling of being nimble, fast and powerful.

Analysis from a gender perspective

When applying Ehrnbergers et al. method of gender driven design, a sliding scale was created to determine how male or female driven the P-40´s design cues are. As presented in the theory section, male related cues are correlated with performance and female related design cues are correlated with helpfulness. In order to illustrate this, a Lamborghini was used to represent a performance driven design, while a Fiat 500 was used to represent a helpfulness driven design. On the scale of 1 to 10, the P-40 has been placed on 6,5.



The main design cues for placing the P-40 at the 6,5 marker from a performance perspective, is the large air intake, multiple exhaust pipes and large engine. The air intake and exhaust indicate that the P-40 has a powerful source of propulsion and is thereby fast. The P-40´s compact and springy composition conveys a certain amount of explosiveness and resilience which are correlated to performance. Furthermore, the aerodynamic shape is cut off vertically in the front which gives a powerful impression resembling a comet.

The main helpfulness driven design cues identified are smooth transitions between surfaces, gentle tapering and the long sweeping lines that can be seen on the fuselage and wings. The smooth transitions evoke a welcoming feeling of friendliness as they seamlessly melt into each other. The gentle tapering and sweeping lines evoke a feeling of thoughtfulness and ease of use. Furthermore, the P-40 does not have any jagged edges or excessive elements protruding from the main body. This relatively minimalistic appearance gives the P-40 an elegant composition which correlates with helpfulness driven design.

With the moderate use of both performance and helpfulness driven design cues, the P-40 appearance is one that engages but still evokes a feeling of caution. It feels accessible and welcoming but at the same time indicates that it is a performance driven piece of engineering. It balances between both extremes and that might be the reason why the P-40´s design has become so iconic.

Analysis from a holistic perspective

Ocke Mannerfelts approach to deciphering and applying design cues share multiple similarities with Karjalainens recognition of explicit and implicit design cues. The similarity is that Mannerfelt extracts a limited amount of visual and emotional design cues from donor to transfer to the subject. The main difference is that Mannerfelt puts them in relation to customer needs and production costs. With this in mind, Mannerfelt transfers design ques that are realistic and stand a fair chance of being produceable.

Based on Mannerfelts method, it is difficult to determine which elements are cost effective to transfer without extensive production knowledge. However, there are several design elements that give the P-40 it’s characteristic appearance. Firstly, the P-40 has long sweeping lines that promote an aerodynamic shape. Both on the fuselage, wings and pilot canape, smooth convex lines are used which minimizes jagged edges. This aerodynamic composition can be compared with a shark that is streamlined to be as efficient as possible. Furthermore, as noted when analyzing the P-40 with Karjalainens design methodology, is the P-40s air intake under the propeller and the exhaust pipes on the sides. These design elements create a contrast to the

otherwise sleek composition. They protrude and give the P-40 a more aggressive appearance that hints on performance and danger.

The final design cue that stands out is the complete composition of the P-40 that creates an emotion of drive and trustworthiness. This emotion can be a construction of prior historical and cinematical portraitations of the P-40, but none the less, still apparent. The large engine to body ratio, the wide shapes that taper off towards the rear and the compact composition enhance the P-40s feeling of being powerful and solid.

Without deeper knowledge of boat production, all of the mentioned elements from the P-40 seem reasonable to be transferred to a boat design from a production and cost perspective.

Color, material and finish

The P-40 has a traditional military color palette. That is to say that the top of the plane is predominantly painted in mat green and brown tones, while the underside is painted in a cold grey. The P-40 was painted in a multitude of colors and camouflages depending on location for duty. However, the colors are in mat finish which minimizes light reflection. The reason for this color and finish choice, is to make the P-40 blend in with its surroundings, regardless if it is seen on the ground or in the air.[20] In addition to the forest green palette, the P-40 has a red nose cone and symbols painted on the entire airplane. As explained in the restrictions, the painted symbols have been disregarded as they are perceived as brand logos rather than design cues.

The P-40´s outer skin is predominantly made from sheet aluminum that is riveted together.[21] The use of sheet aluminum gives the P-40 a tight and smooth appearance which is experienced as elegant. Other material factors that contribute to the elegant feeling is minimalistic use of other materials. Besides the glass canopy, sheet aluminum is the only material seen from the outside.

[20] Guttman, Curtiss P-40 Warhawk: One of WWll`s most famous fighters.
[21] Guttman, Curtiss P-40 Warhawk: One of WWll`s most famous fighters.



[22]

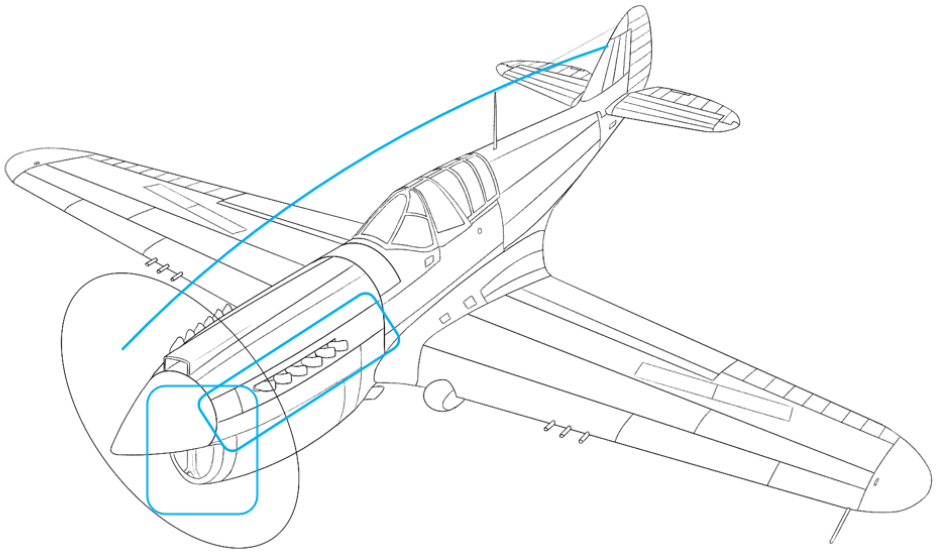
[22] Image nr 4. Reddit, P-40 blue sky.

Implementation

Explicit design cues & Mannerfelt

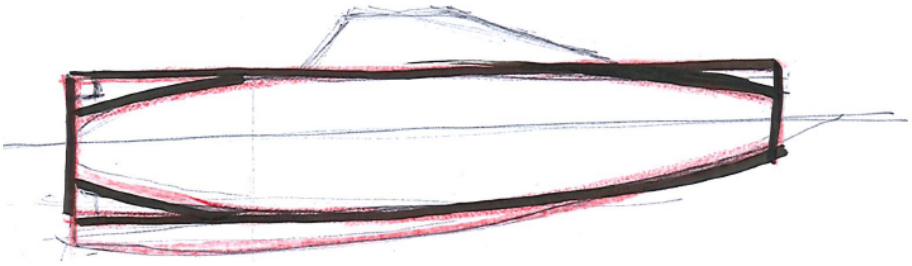
When designing the Mallard, a center console boat inspired by the P-40, each theoretical method was used individually to analyze the airplane. However, once the design process began, it became apparent how well the theories interconnected with each other and created synergy effects.

Starting with Mannerfelts theory of extracting three fundamental lines from the donor product, it had many similarities to Karjalainens therory of explicit and implicit design cues. Therefore, three explicit design cues from the P-40 were chosen that stood out. The ones chosen were convex springy lines, large protruding air intake and the multiple stumpy exhaust pipes.

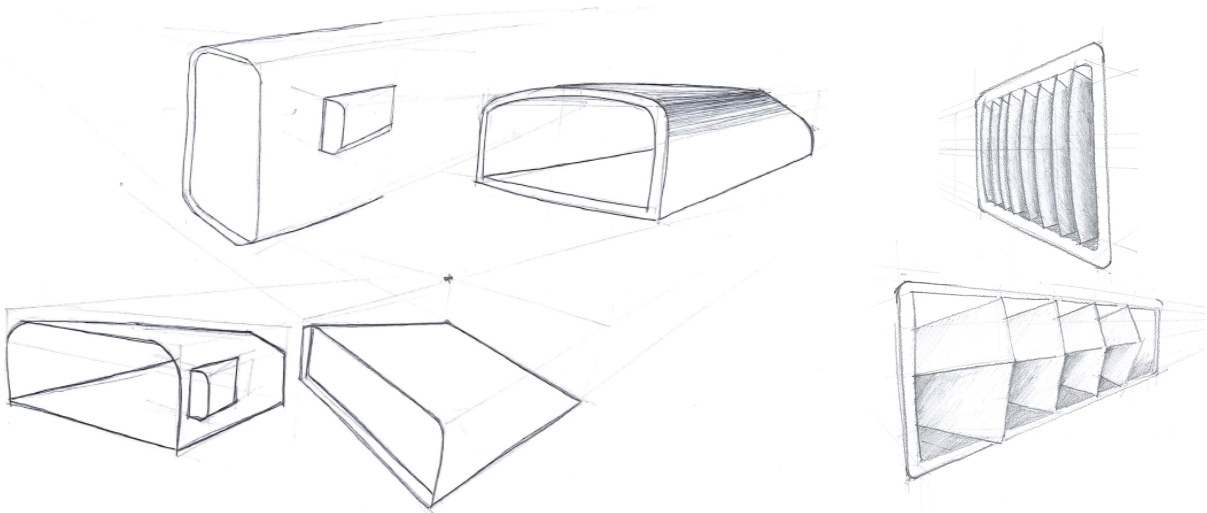


In order to recreate the P-40´s plum appearance, convex lines were used. The convex lines were placed both horizontally and vertically on the Mallards components like the hull, center console and the transom. The convex lines immediately give the Mallard a plum and aerodynamic feeling that is experienced with the P-40. This is especially evident when

looking at the Mallard from a profile view where the lines continually flow from the bow to the stern.



As both the P-40 and the Mallard are powered by a gasoline engine, an air intake is needed to feed the combustion engine. This mutual necessity for air, makes it natural to adopt the characteristic air intake element from the P-40 and apply it to the Mallard. Although the P-40´s engine and intake are placed in the front, placing the design cue on the back of the boat makes sense as there is where the boat's engine is located. The boat's placement of the air intake adopts the same design cue from the P-40. That is to say, the sleek overspanning line of the fuselage is broken off with a large and protruding air intake.



Finally, the last explicit design cues transferred from the P-40 to the boat, are the multiple and stumpy exhaust pipes. Once again, as both the P-40 and the Mallard have combustion engines, this transfer could have been straight forward. However, as the intention is to create a boat that resembles the P-40 and not copies it, the side mounted exhaust pipes were replaced with

side mounted air vents on the Mallard instead. When designing these air vents, they were not designed from scratch. Instead, existing air from existing boats were used. These production boat vents coincidently share the same design cues as the exhaust pipes from the P-40. They both have several backward angled slots that are aligned behind each other. Furthermore, they relate with Mannerfelts theory of using components that are cost efficient and realistic.

Implicit design cues

As mentioned in the analysis section, the most prominent implicit design cues identified, were the sensation of the P-40 being nimble, moderatly fast and powerful.

The design traits that generated this sensation are the P-40´s square layout when seen from a top view. The P-40´s shape is a symmetrical cross, that conveys a feeling of it being nimble and quick. This symmetrical layout, in combination with a short wingspan means that it's turning radius is small, corolating with nimbleness. The P-40 can be compared with a squared go-cart which is more nimble around corners than a long car with an extended wheelbase.



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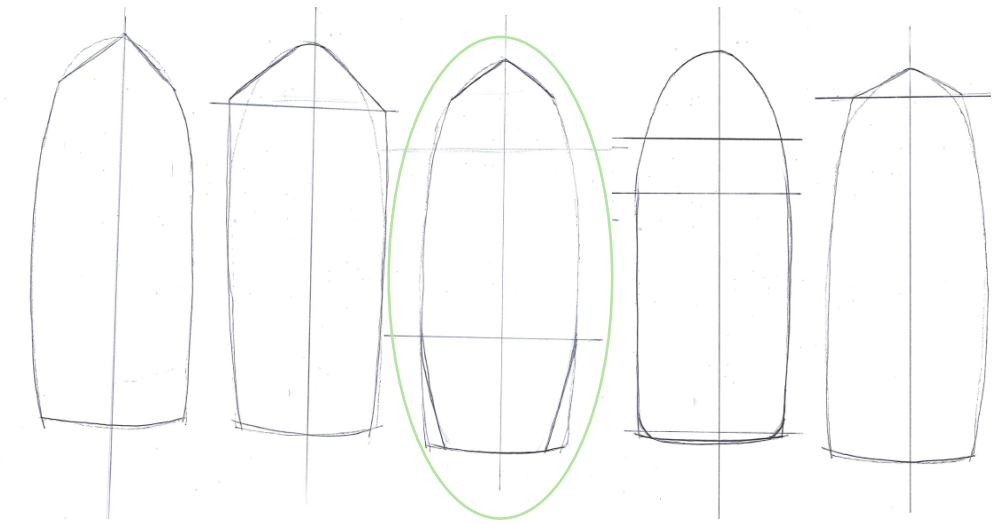
24

A second noticable trait of the the P-40, is its massive engine and propeller compartment. This compartment covers nearly one third of the plane's overall length and instigates a sensation of the P-40 being fast and powerful. Other design cues that also convey the senastion of speed are long sweeping lines and smooth surfaces with few detalis that protrude from the

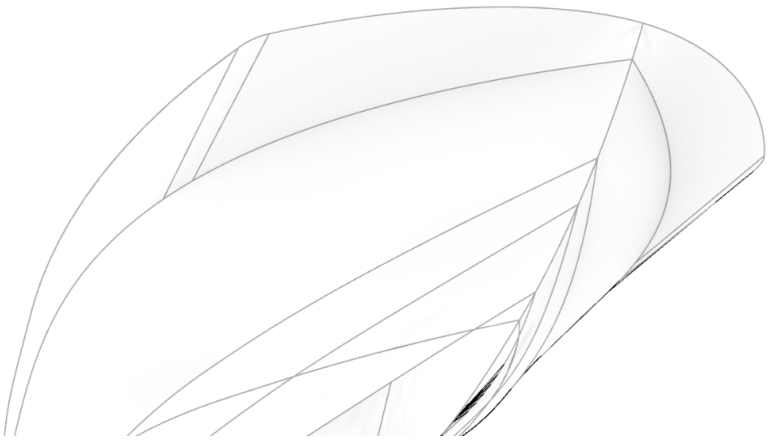
[23] Image nr 5. Volts, Go-kart.
[24] Image nr 6. Blacklimoride, Lincoln Navigator.

areodynamic shape. Almost like a shark, nothing unessessary that increases drag sticks out from it's body.

In order to transfer the implicit design cues of nimbleness and power, the boat was designed with simplicity in mind. First of all, the over all shape of the Mallard was made short and compact. Although boats are traditionally longer than wider in order to move efficently through water, the Mallard was designed as compact as possible. The Mallard was also designed with a two component hull. This means that the lower part of the hull, which is in contact with water, is sharp in the bow. This lets the Mallard cut through waves in an effcient way. However, the upper part of the hull, which has no contact with water, is made blunt to mimique the fusalage front of the P-40 which is vertically straight. This silhouette can be dircribed as an american football with it's front cone removed. With this said, the bow of the Mallard is rounded-off which breaks the otherwise sweaping hull line.



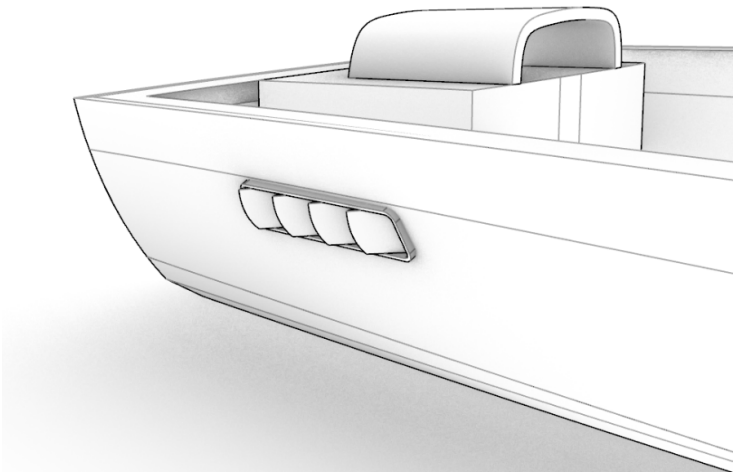
The intention of splitting up the hull in two sections, is to convey a quick yet nimble sensation equvilant to the senation generated by the P-40. The lower part of the hull is responsible for the feeling of speed and power, while the top section of the hull is responsible for conveying a feeling of nimbleness.



Gender driven design cues

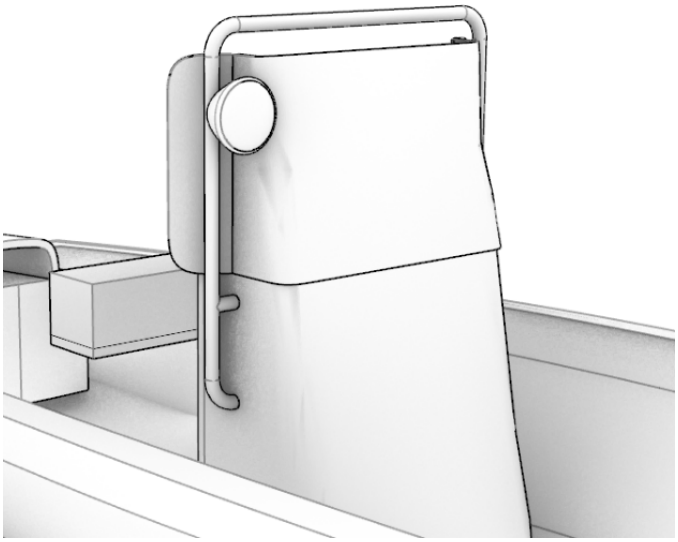
In addition to the explicit and implicit design cues chosen, the Mallard was designed to achive the same level of preformance vs helpfullness esthetics as the P-40´s rating of 6,5. To do so, decisions were taken to make the Mallard minimalistic in terms of seating arrangements, practical solutions and details. The reason for this is that the P-40 does not have any unnecessary features, especially not features that would add extra wind resistance. With this in mind, the boat was design around the helm or center console where the boat is driven from. The center console is placed in the middle of the boat for the captain to have full control of everything happening abord.

Instead of makeing multiple preformance-driven design cues throuhout the entire Mallard, the P-40´s design approch was adopted were prefromance indicators are concentrated around the engine compartment. With this said, the Mallard was equipped with a large air intake that breaks the otherwise smooth silhouette of the boat. The engines ventilation vents were made oversized and placed on each side of the hull. The ventilation vents were designed with diagonal slanting fins which imply direction and speed. The intention with the ventilation vents are to mimic the P-40´s exahust pipes which also have multiple backwords slanting pipes.



Finally, in order to accentuate the Mallards preformance resemblance to the P-40, it was designed with a duo-propeller instead of the conventional single propeller. This preformance associated propeller gives the Mallard a subtle yet distinct preformance indicator which differntiates it from a conventional recreational boat. It is a detail that higlights the Mallard's preformance heritage without being abnoctious.

The Mallard's helm was also designed to replicate the P-40´s 6,5 value on the preformance vs helpfullness scale. The helmstation is minimalistic with only the essential stearing components and gauges. It is designed around the captain but also allows a passanger to stand next to him if needed. Instead of using traditonal seats at the helm, a standing support-seat is used to portray a sensation of a utility boat with an active and sporty driving position. By standing instead of sitting the captain can easily move and have a broader vision range. These factors make the Mallard feel slightly more preformace driven than helpfulllness driven which was the desierd effect.



To finish off the helmstation, foldable side windscreens and an integrated search light were added as final details. These details have a practical aspect which corrolates with the p-40, but they also instigate a moderate amount of performance. They are just enough to portray that the Mallard is designed for active use without beeing over functional.

Color, material and finish

The P-40´s CMF was primaraly designed to aid the plane in war situations. Although functional, the P-40´s camoflage pattern, aluminium bodypanels and matt finish, give the airplane a spartan feel. In contrast to the P-40, the Mallard's objetive is to be experianced as a stylish, recreational boat for sunny days. In order to achieve this differenciation, the Mallard's hull is painted in a dark forest green a.k.a British racing green. This tone of color creates resemblance to the P-40 and the single main color creates sense of calmness closely related to elegance. Furthermore, the use of a single hull color instead of a pattern, moves the focus from the boats color scheme to the total design composition of the boat.



[25]

To further strengthen the feeling of elegance, the Mallard was given a glossy finnish. Besides having an elegant effect, there is a maritime traditon that encourages varnished boat hulls. Other elements that also were made glossy instead of matt include railings, the stearing wheel and the searchlight. Once again this choice was taken to create elegance and maritime accoiation.

The interior of the Mallard was paitned in a creamy off-white. This white tone has an aged feeling more associated with elegance than a crispy white. The creamy white furthermore goes well with the tan upholsteries and the racing green. In order to break off the the greens and browns, a red waterline was added. The red waterline is inspired from the P-40´s red nose cone and creates a resemblance between the two products without beeing overstated. Although most finishing choices on the Mallard contradict the ones from the P-40, they are still considered appropriate as they create resemblance but still give the Mallard it's own personality.

Finally, the Mallard was designed in fiberglass and not sheet aluminium as the P-40. The reason for fiberglass is its strength, moldabilty and low cost.

[25] Image nr 7. Pilotweb, P-40 skull decal.

Results

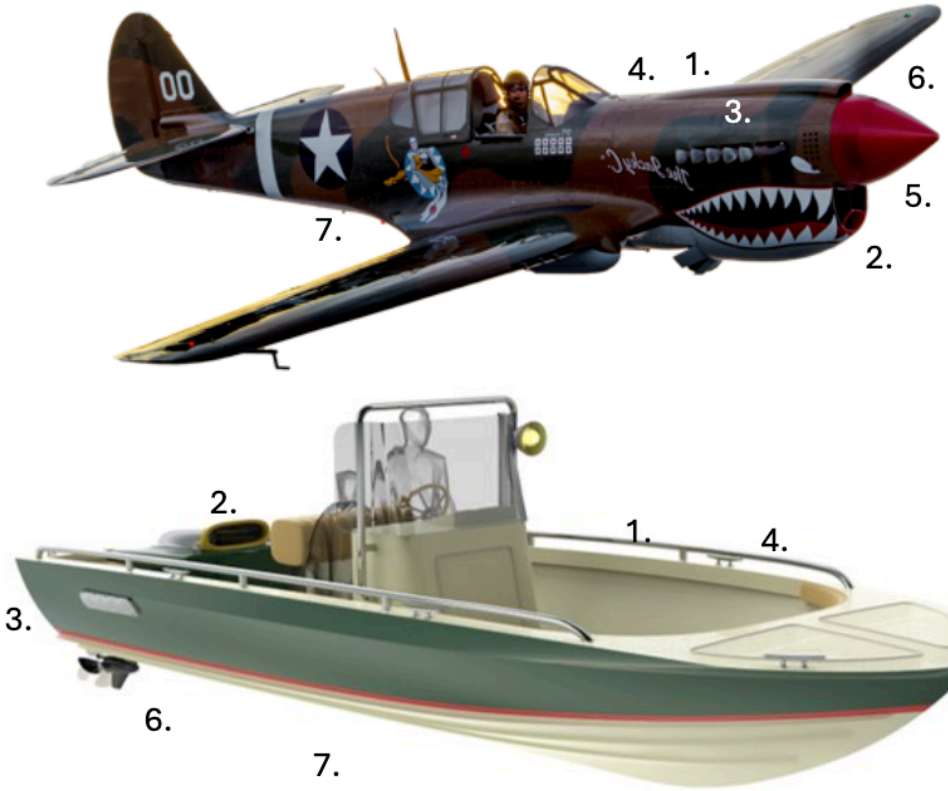
The objective of this thesis was to analyze the P-40 through the lens of several design theories and thereafter design a boat based on the identified design cues. With this in mind, the goal with Karjalainen's design driven theory, was to transfer both explicit and implicit design cues. The explicit design cue chosen were convex springy lines, the protruding air intake as well as multiple stumpy exhaust pipes. The implicit design cues chosen to be transferred were the sensation of being nimble, moderately fast and powerful. As for Ehrnberger's et al. gender driven design theory, the goal was to recreate the P-40´s 6,5 position on the performance vs helpfulness scale. Finally, the goal with applying Mannerfelt's holistic design approch on the Mallard, was to make it resemble the P-40, but also have in consideration cost and production limitations.



The P-40´s convex lines were applied on multiple elements like the hull, air intake and railing. Most notable is the footprint of the hull when seen from above. It has a characteristic convex outline that resembles an american football with it's extremities removed. This aerodynamic

shape with blunt extremities is one of the P-40´s fundamental design cues and creates a clear similarty between boat and airplane. Not only does it create resemblence, but it also gives the Mallard a sensation of being nimble and explosive.

The P-40's distinct air intake and multiple exahust pipes were also transferred to the Mallard. As both boat and air plane have engines, this transfer was experianced as natural. However, these elements not only add to the explicit design cues, they also give the Mallard an implicit sensation of being powerful. They indicate that the Mallard does have a higher level of performance without beeing a racing boat. This level of moderate performance not only relates to Karjalainen's implicit design cues, but it also strengthens it's placement of 6,5 on Ernberger's et al. preformance vs helpfulness scale.

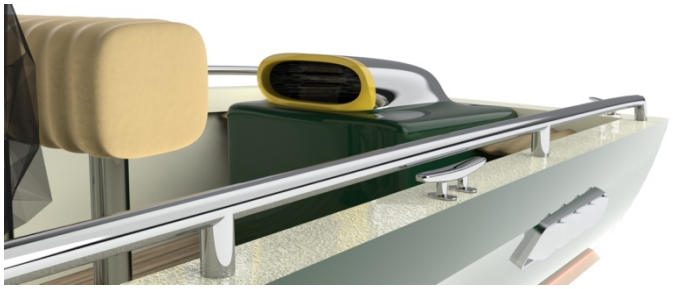


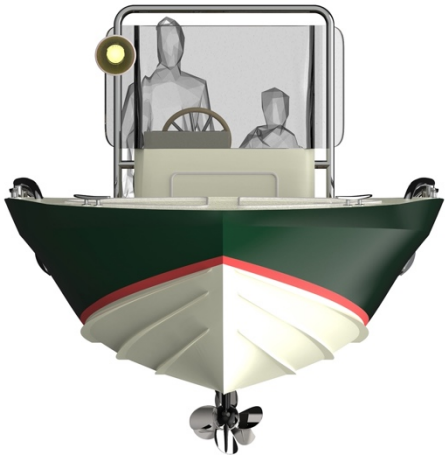
1. Convex lines.
2. Air intake.
3. Exhaust/ vents.
4. Nimble.
5. Moderately fast.
6. Powerful.
7. Heritage color palette.

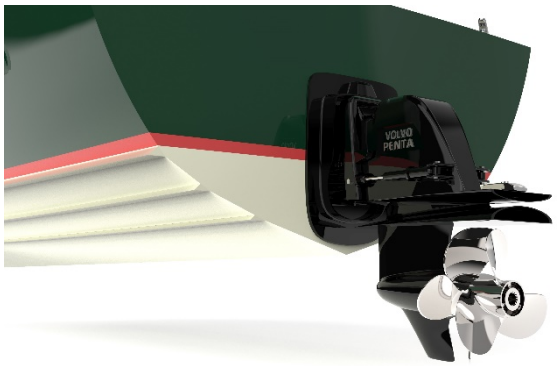
Additional elements that give the Mallard a sensation of beeing moderately fast, is the perpendicular and squared center console. This square center console breaks off the overall aerodynamic sensation and gives the Mallard more air resistance. This blunt shape furthermore reoccurs on details such as the captain's seat and the air intake box. These details

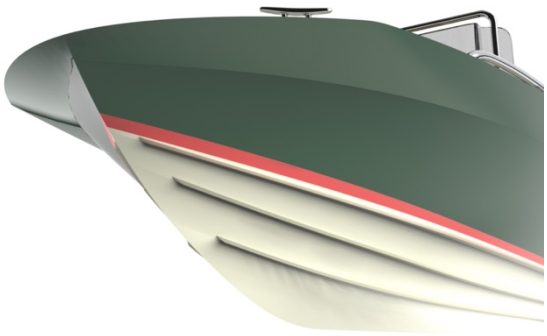
strengthen the feeling of the Mallard just beeing moderately fast and not an aerodynamic speed boat.

In order for the Mallard to feel as nimble as the P-40, the Mallard has an uncluttered design that revolves around the center console. This composition built around the center console generates a nimble and distinct appearance adopted from the P-40. Being designed around the center console gives the Mallard a natural pivot-point close to the center which has a strong correlation to balance and nimbleness. Furthermore, the combination of a sharp lower-hull and a blunt top-hull give the Mallard a semi-nimble esthetic. The use of a dual-propeller was added as a performance indicator to boost the sensation of speed and power. However, additional design choices taken to balance the Mallards helpfulness value were few angular edges, a blunt bow and a round search light.









Conclusion

When deciphering and applying the P-40´s design cues to the Mallard, all three design theories explored were functional. Karjalainens visually driven design theory is extensive by taking into account both the explicit attributes of the P-40, but also the implicit emotional sensations experienced. Dividing design cues into explicit and implicit gives the analysis depth but also gives the designer flexibility to create an overall sensation that mimics the donor product without copying every design element. For example, the Mallard can be associated with the P-40 without having wings or a cockpit canopy.

Ehrnbergers et al. gender driven design theory was experienced as a radical but efficient design method. Instead of transferring explicit design cues, the theory focuses on transferring the same level of performance vs helpfulness indicators. This flexibility of beeing able to discard the use of specific design cues, gives the designer the freedom to incorporate new design elements that otherwise would be considered unsuitable. With this said, gender driven design theory creates space for creativity and innovation in the design process.

Mannerfelts holistic approach to transferring design cues is more ridged but also easier to relate to. His design theory encourages the designer to limit themselves to three primary design cues but also have customer- and production limitations in mind. These relatively ridged guidelines narrow down the design options and force the designer to focus on fundamental design cues. Although this approach can be considered limiting, it is the limitations that make the theory relatable and useful in real life.

There were two main difficulties encountered when transferring the P-40´s design cues. The first difficulty was the desire to make the Mallard esthetically pleasing. With this said, it was challenging to stay true to the design theories as the results were not always what was expected. The solution to this problem was to take a step back and have faith in the process.

The second difficulty encountered was the impulse to recreate a copy of the P-40 in boat form. This means that the Mallard occasionally became overworked and lost its own identity. The solution to this problem was to keep the goal of this thesis in mind. The goal is not to recreate an exact copy of the P-40, but instead use the P-40 as a source of inspiration.

Finally, the main conclusion reached from implementing existing design theory when transferring design cues from the P-40 to the Mallard, was the importance of experience and talent. As Mannerfeldt expressed, there is no exact science or exact set of rules to follow in order to transfer design cues. Although theory is an essential start point, a designer's experience and talent are crucial to achieve a successful adoption and implementation of design cues.



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